

Modem diag procedure

Check the following steps in order to troubleshoot the **P100 Sierra Wireless Modem**

Before start check which version of teensy we are running: should be at least v0.1.12.

```
1 python -c 'from teensycmds.teensycmds import TeensyBoard; print(TeensyBoard().cmd("ping"))'
```

- Check the internet.py in crontab to check the modem is being enabled

```
1 [VOE:EC-NPC] root@p100-ify:/# grep internet /etc/crontab
2 #Ansible: internet
3 * * * * * root /usr/bin/python /root/p100_tools/internet/internet-p100.py
```

- Check dmesg in order to see the modem is on and detected correctly (only important parts were pasted in following snippet, you should see more usb entries:

```
1 dmesg -T |grep -iE 'LTE-A|cdc|modem'
```

A wider version will be this one

```
1 [VOE:EC-NPC] root@p100-ify:/# dmesg |grep usb
2 usb 1-3.3: new high-speed USB device number 7 using xhci_hcd
3 usb 1-3.3: New USB device found, idVendor=1199, idProduct=9091
4 usb 1-3.3: New USB device strings: Mfr=1, Product=2, SerialNumber=3
5 usb 1-3.3: Product: Sierra Wireless EM7565 Qualcomm\xffffffc2\xffffffae\xffffffae
6 Snapdragon\xffffffe2\xffffff84\xffffffa2\xffffff84\xffffffa2 X16 LTE-A
7 usb 1-3.3: Manufacturer: Sierra Wireless, Incorporated
8 usb 1-3.3: SerialNumber: UF94267130041546
9 usbcore: registered new interface driver qcserial
10 usbserial: USB Serial support registered for Qualcomm USB modem
11 usb 1-3.3: Qualcomm USB modem converter now attached to ttyUSB3
12 usb 1-3.3: Qualcomm USB modem converter now attached to ttyUSB4
13 usbcore: registered new interface driver cdc_ncm
14 usbcore: registered new interface driver cdc_wdm
15 cdc_mbim 1-3.3:1.0 wwan0: register 'cdc_mbim' at usb-0000:00:14.0-3.3, CDC MBIM,
42:91:cc:a1:bf:57
16 usbcore: registered new interface driver cdc_mbim
```

- How to detect correct AT port:

```
1 readlink /sys/bus/usb-serial/devices/ttyUSB*|sed -ne 's,.*3/tty,tty,p'
```

- **Important** in some situations you should not be able to write anything on terminal, to troubleshoot this, check AT port (precedent snippet, lines 10/11), in this particular case ttyUSB3 and enable “local echo” and try again:

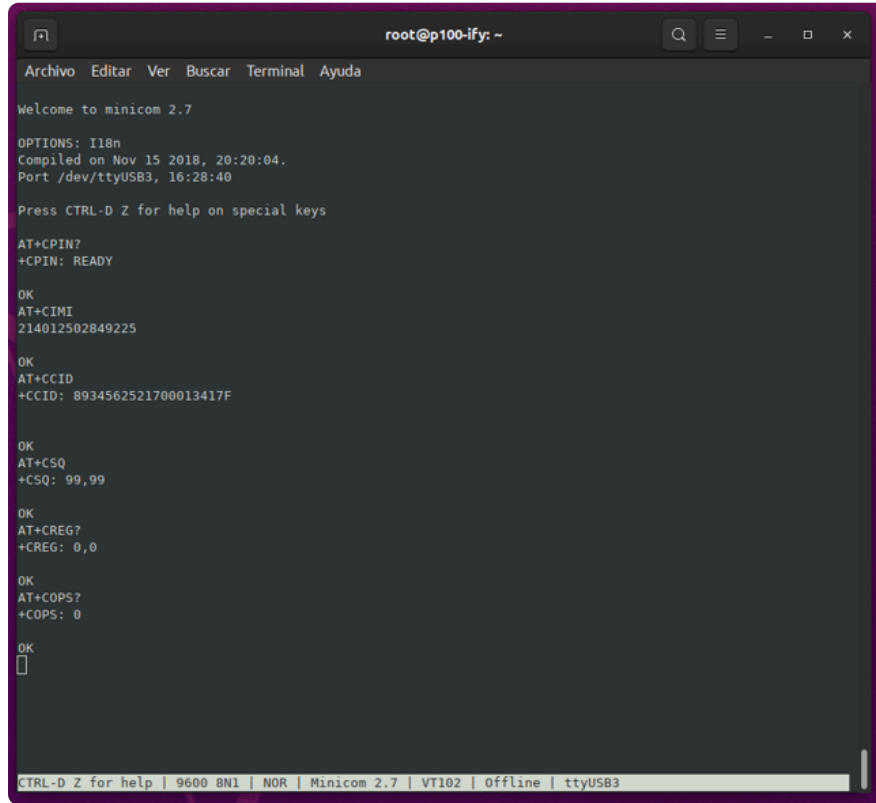
```
+-----[configuration]-----+
| Filenames and paths          |
| File transfer protocols      |
| Serial port setup           |
| Modem and dialing          |
| Screen and keyboard       |
| Save setup as dfl         |
| Save setup as...            |
| Exit                         |
| Exit from Minicom           |
+-----+
```

```
+-----[Screen and keyboard]-----+
| A - Command key is          : ^D
| B - Backspace key sends    : BS
| C - Status line is         : enabled
| D - Alarm sound             : Yes
| E - Foreground Color (menu): WHITE
| F - Background Color (menu): BLACK
| G - Foreground Color (term): WHITE
| H - Background Color (term): BLACK
| I - Foreground Color (stat): WHITE
| J - Background Color (stat): BLACK
| K - History Buffer Size     : 2000
| L - Macros file             : .macros
| M - Edit Macros             |
| N - Macros enabled          : Yes
| O - Character conversion    :
| P - Add linefeed            : No
| Q - Local echo              : Yes
| R - Line wrap               : No
| S - Hex Display             : No
| T - Add carriage return    : No
| Change which setting? (Esc to exit)
+-----+
```

```
+-----[configuration]-----+
| Filenames and paths          |
| File transfer protocols      |
| Serial port setup           |
| Modem and dialing           |
| Screen and keyboard       |
| Save setup as dfl         |
| Save setup as...       |
| Exit                         |
| Exit from Minicom           |
+-----+
```

Important bis: If you see that once passed some you obtain doubled characters from the terminal, it's time to disable “Local echo” and leave it as “NO” as before.

- Now you can point to /dev/ttyUSB3 in minicom and start sending some commands and wait for output:



```
root@p100-ify: ~
Archivo  Editar  Ver  Buscar  Terminal  Ayuda

Welcome to minicom 2.7

OPTIONS: I18n
Compiled on Nov 15 2018, 20:20:04.
Port /dev/ttyUSB3, 16:28:40

Press CTRL-D Z for help on special keys

AT+CPIN?
+CPIN: READY

OK
AT+CIMI
214012502849225

OK
AT+CCID
+CCID: 8934562521700013417F

OK
AT+CSQ
+CSQ: 99,99

OK
AT+CREG?
+CREG: 0,0

OK
AT+COPS?
+COPS: 0

OK
█

CTRL-D Z for help | 9600 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyUSB3
```

The commands explained:

```
1 //Check sim is unlocked
2 AT+CPIN?
3 //Check IMSI (International Mobile Subscriber Identity)
4 AT+CIMI
5 //AT+CCID AT command is used to read the ICCID from the SIM.
6 AT+CCID
7 //Check the signal of the modem <RSSI,BER>, RSSI = 99 = no signal
8 AT+CSQ
9 //Check if the SIM is being registred in the some network
10 AT+CREG?
11 //Check which network is being used by SIM if registred
12 AT+COPS?
13
```

- Scriptable test

- Compile atinout & copy to /sbin:

```
1 git clone git://git.code.sf.net/p/atinout/code atinout
```

- Launch following command

```
1 | (echo AT+CPIN?; echo AT+CIMI; echo AT+CCID; echo AT+CSQ; echo AT+CREG?; echo AT+COPS?) \  
2 | atinout - /dev/ttyUSB3 -
```

Known problems

1-Several reboots and then DLM

```
142.676487] usb 1-3.3: New USB device strings: Mfr=1, Product=2, SerialNumber=3  
142.676491] usb 1-3.3: Product: Sierra Wireless EM7565 Qualcomm\xffffffc2\xfffffffae\xfffffffae Snapdragon\xffffffe2\xffffff84\xfffffffa2\xffffff84\xfffffffa2 X16 LTE-A  
142.676494] usb 1-3.3: Manufacturer: Sierra Wireless, Incorporated  
142.676497] usb 1-3.3: SerialNumber: UF93467377041507  
142.689031] usbcore: registered new interface driver qcserial  
142.689075] usbserial: USB Serial support registered for Qualcomm USB modem  
142.689586] usb 1-3.3: Qualcomm USB modem converter now attached to ttyUSB3  
142.689607] usb 1-3.3: Qualcomm USB modem converter now attached to ttyUSB4  
142.700179] usbcore: registered new interface driver cdc_ncm  
142.703360] usbcore: registered new interface driver cdc_wdm  
142.727591] cdc_mbm 1-3.3:1.0 wwan0: register 'cdc_mbm' at usb-0000:00:14.0-3.3, CDC MBIM, 02:73:b8:54:04:f8  
142.727640] usbcore: registered new interface driver cdc_mbm  
152.628123] usb 1-3.3: USB disconnect, device number 7  
152.628227] cdc_mbm 1-3.3:1.0 wwan0: unregister 'cdc_mbm' usb-0000:00:14.0-3.3, CDC MBIM  
164.349081] usb 1-3.3: new high-speed USB device number 8 using xhci_hcd  
164.438488] usb 1-3.3: New USB device found, idVendor=1199, idProduct=9091  
164.438498] usb 1-3.3: New USB device strings: Mfr=1, Product=2, SerialNumber=3  
164.438505] usb 1-3.3: Product: Sierra Wireless EM7565 Qualcomm\xffffffc2\xfffffffae\xfffffffae Snapdragon\xffffffe2\xffffff84\xfffffffa2\xffffff84\xfffffffa2 X16 LTE-A  
164.438510] usb 1-3.3: Manufacturer: Sierra Wireless, Incorporated  
164.438516] usb 1-3.3: SerialNumber: UF93467377041507  
164.460761] cdc_mbm 1-3.3:1.0 wwan0: register 'cdc_mbm' at usb-0000:00:14.0-3.3, CDC MBIM, 02:73:b8:54:04:f8  
164.461525] usb 1-3.3: Qualcomm USB modem converter now attached to ttyUSB3  
164.462094] usb 1-3.3: Qualcomm USB modem converter now attached to ttyUSB4  
174.390244] usb 1-3.3: USB disconnect, device number 8  
174.390332] cdc_mbm 1-3.3:1.0 wwan0: unregister 'cdc_mbm' usb-0000:00:14.0-3.3, CDC MBIM  
186.111326] usb 1-3.3: new high-speed USB device number 9 using xhci_hcd  
186.200997] usb 1-3.3: New USB device found, idVendor=1199, idProduct=9091  
186.201006] usb 1-3.3: New USB device strings: Mfr=1, Product=2, SerialNumber=3  
186.201013] usb 1-3.3: Product: Sierra Wireless EM7565 Qualcomm\xffffffc2\xfffffffae\xfffffffae Snapdragon\xffffffe2\xffffff84\xfffffffa2\xffffff84\xfffffffa2 X16 LTE-A  
186.201018] usb 1-3.3: Manufacturer: Sierra Wireless, Incorporated  
186.201024] usb 1-3.3: SerialNumber: UF93467377041507  
186.223792] cdc_mbm 1-3.3:1.0 wwan0: register 'cdc_mbm' at usb-0000:00:14.0-3.3, CDC MBIM, 02:73:b8:54:04:f8  
186.224531] usb 1-3.3: Qualcomm USB modem converter now attached to ttyUSB3  
186.225610] usb 1-3.3: Qualcomm USB modem converter now attached to ttyUSB4  
196.152495] usb 1-3.3: USB disconnect, device number 9  
196.152593] cdc_mbm 1-3.3:1.0 wwan0: unregister 'cdc_mbm' usb-0000:00:14.0-3.3, CDC MBIM  
208.129536] usb 1-3.3: new high-speed USB device number 10 using xhci_hcd  
208.218836] usb 1-3.3: New USB device found, idVendor=1199, idProduct=9091  
208.218846] usb 1-3.3: New USB device strings: Mfr=1, Product=2, SerialNumber=3  
208.218853] usb 1-3.3: Product: Sierra Wireless EM7565 Qualcomm\xffffffc2\xfffffffae\xfffffffae Snapdragon\xffffffe2\xffffff84\xfffffffa2\xffffff84\xfffffffa2 X16 LTE-A  
208.218859] usb 1-3.3: Manufacturer: Sierra Wireless, Incorporated  
208.218864] usb 1-3.3: SerialNumber: UF93467377041507  
208.241488] cdc_mbm 1-3.3:1.0 wwan0: register 'cdc_mbm' at usb-0000:00:14.0-3.3, CDC MBIM, 02:73:b8:54:04:f8  
208.242302] usb 1-3.3: Qualcomm USB modem converter now attached to ttyUSB3  
208.242707] usb 1-3.3: Qualcomm USB modem converter now attached to ttyUSB4  
217.914632] usb 1-3.3: USB disconnect, device number 10  
217.914730] cdc_mbm 1-3.3:1.0 wwan0: unregister 'cdc_mbm' usb-0000:00:14.0-3.3, CDC MBIM  
229.635659] usb 1-3.3: new high-speed USB device number 11 using xhci_hcd  
229.724938] usb 1-3.3: New USB device found, idVendor=1199, idProduct=9091  
229.724947] usb 1-3.3: New USB device strings: Mfr=1, Product=2, SerialNumber=3  
229.724953] usb 1-3.3: Product: Sierra Wireless EM7565 Qualcomm\xffffffc2\xfffffffae\xfffffffae Snapdragon\xffffffe2\xffffff84\xfffffffa2\xffffff84\xfffffffa2 X16 LTE-A  
229.724958] usb 1-3.3: Manufacturer: Sierra Wireless, Incorporated  
229.724963] usb 1-3.3: SerialNumber: UF93467377041507  
229.737423] cdc_mbm 1-3.3:1.0 wwan0: register 'cdc_mbm' at usb-0000:00:14.0-3.3, CDC MBIM, 02:73:b8:54:04:f8  
229.738532] usb 1-3.3: Qualcomm USB modem converter now attached to ttyUSB3  
229.739203] usb 1-3.3: Qualcomm USB modem converter now attached to ttyUSB4  
239.676847] usb 1-3.3: USB disconnect, device number 11  
239.676958] cdc_mbm 1-3.3:1.0 wwan0: unregister 'cdc_mbm' usb-0000:00:14.0-3.3, CDC MBIM  
251.397815] usb 1-3.3: new high-speed USB device number 12 using xhci_hcd  
251.487186] usb 1-3.3: New USB device found, idVendor=1199, idProduct=9091  
251.487196] usb 1-3.3: New USB device strings: Mfr=1, Product=2, SerialNumber=3  
251.487201] usb 1-3.3: Product: Sierra Wireless EM7565 Qualcomm\xffffffc2\xfffffffae\xfffffffae Snapdragon\xffffffe2\xffffff84\xfffffffa2\xffffff84\xfffffffa2 X16 LTE-A  
251.487206] usb 1-3.3: Manufacturer: Sierra Wireless, Incorporated  
251.487211] usb 1-3.3: SerialNumber: UF93467377041507  
251.509668] cdc_mbm 1-3.3:1.0 wwan0: register 'cdc_mbm' at usb-0000:00:14.0-3.3, CDC MBIM, 02:73:b8:54:04:f8  
251.510414] usb 1-3.3: Qualcomm USB modem converter now attached to ttyUSB3  
251.510937] usb 1-3.3: Qualcomm USB modem converter now attached to ttyUSB4  
261.439029] usb 1-3.3: USB disconnect, device number 12  
261.439130] cdc_mbm 1-3.3:1.0 wwan0: unregister 'cdc_mbm' usb-0000:00:14.0-3.3, CDC MBIM  
262.406924] usb 1-3.3: new high-speed USB device number 13 using xhci_hcd  
262.497803] usb 1-3.3: New USB device found, idVendor=1199, idProduct=9090  
262.497813] usb 1-3.3: New USB device strings: Mfr=1, Product=2, SerialNumber=3  
262.497819] usb 1-3.3: Product: Sierra Wireless EM7565 Qualcomm\xffffffc2\xfffffffae\xfffffffae Snapdragon\xffffffe2\xffffff84\xfffffffa2\xffffff84\xfffffffa2 X16 LTE-A  
262.497825] usb 1-3.3: Manufacturer: Sierra Wireless, Incorporated  
262.497830] usb 1-3.3: SerialNumber: UF93467377041507  
262.499157] usb 1-3.3: Qualcomm USB modem converter now attached to ttyUSB3  
VOE:EC-NOY root@p100-ify:~#
```

